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.Net ViewState -AnquanKER - Author not stated, anquanke.com.

- Title in English: .Net deserialization: exploiting ViewState - AnquanKER
- Published: 2020-11-05
- Original: <https://www.anquanke.com/post/id/221630>
- Preserved from: <https://www.anquanke.com/post/id/221630> (preserved-copy) on 2026-08-04
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.Net deserialization: exploiting ViewState

Views **441487**

Published: 2020-11-05 10:30:09

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Among .NET related vulnerabilities, ViewState is a regular guest. ViewState has shown up in both Exchange CVE-2020-0688 and SharePoint CVE-2020-16952. In fact ViewState is not a vulnerability in itself; it is only that ASP.NET uses ObjectStateFormatter to serialize and deserialize when it generates and parses ViewState. Although the serialized data is then encrypted and signed, once the algorithms and keys used for encryption and signing leak, we can disguise an ObjectStateFormatter deserialization payload as a normal ViewState and trigger the ObjectStateFormatter deserialization vulnerability.

The algorithms and keys used to encrypt and sign the serialized data are stored in web.config. Exchange 0688 happened because every installation used the same default key, while Sharepoint 16952 happened because web.config leaked.

The .NET deserialization tool of choice, ysoserial.net, has a plugin for ViewState. Its main job is to use a leaked algorithm and key to forge the encryption and signature of a ViewState and trigger the ObjectStateFormatter deserialization vulnerability. But we should not be satisfied with just using the tool, so I deliberately analysed the ViewState encryption and signing process and wrote this article, to understand the tool completely.

I am new to .NET, so mistakes and omissions in this article are unavoidable; corrections are much appreciated.

1. Debugging .Net Framework

1.1 .Net source code

For friends who are new to .Net deserialization, or even new to C#, having a comfortable and convenient debugging environment is really important. Here I will briefly introduce how to do low-level debugging of the .net framework.

.Net Framework has been [open sourced](#) by Microsoft; you can download the source code from the official website or browse it online directly. The versions currently open sourced include .Net 4.5.1 to 4.8. But note that although Microsoft open sourced the .Net source code and the corresponding VS project files, they can only be used for browsing the code, not for compiling, because important components (including xaml files and resource files) are missing.

1.2 Debugging

The official Microsoft documentation explains how to use VS to debug the .Net source. The principle is roughly to single step using pdb plus source code. But after actually trying it, I found that not every .net assembly file has a complete pdb file; the pdb of some assemblies has no source information. In other words, only some assemblies can be single stepped in vs.

See the following links for details: <https://referencesource.microsoft.com/setup.html>

The list of assemblies that support source debugging is: <https://referencesource.microsoft.com/indexedpdbs.txt>

After giving up on debugging with vs, I found that dnspy can also be used for low-level .net debugging. dnspy is an open source .Net decompiler; compared with the classic tool Reflector, it can not only decompile but also debug directly with the help of decompilation. The github link for dnspy is [here](#). You can download the source and compile it, or download a prebuilt version directly, but be careful to satisfy the .net framework version it requires.

Set the environment variable COMPLUS_ZapDisable=1

Why set this environment variable? To disable the use of all NGEN images (*.ni.dll).

Suppose the IIS service is installed on your windows server and a website is running on it. Open that website with a browser; this makes IIS create a worker process in the background to run the website. Now use process explore to look at the dlls loaded by the w3wp.exe process, and you will find that every assembly has a .ni suffix. System.Web.dll has become System.Web.ni.dll, and the description of that dll even says "System.Web.dll". This is in fact the optimized version of the .Net code being used.

Set the environment variable COMPLUS_ZapDisable=1 and restart windows (you must restart, because only restarting the IIS service applies the new environment variable we set). Open the website with ie again, then use Process explore to look at w3wp.exe, and you will find that the assemblies loaded by the website worker process have changed back to the familiar System.Web.dll.

Note 1: after setting the environment variable you must restart

Note 2: if you cannot find w3wp.exe, run process explore as administrator.

Debugging with dnspy

First we use process explore to check where the assemblies loaded by `w3wp.exe` are located, because your system may have several versions of .Net installed, or .Net of different bitness. If you open the wrong assembly in dnspy, then when you set a breakpoint on it you will be told: cannot break at this breakpoint because that module is not loaded.

Choose the 32 bit or 64 bit dnspy (matching the process being debugged) and start it as administrator. Pick any assembly, for example System.Web.dll, open it and see whether the path written on its first line is the same as the assembly loaded by the target process:

If it is not the same, use File -> Close All at the top left, then File -> Open List, and pick a suitable version of .Net from it.

Then use Debug -> Attach to Process at the top and choose `w3wp.exe`. If there are several processes, we can identify it by its process id. So how do we decide which process is the one we want? There are many ways. You can use process explorer to look at the start command of `w3wp.exe` and see which one is the worker process running the target website. Or, start cmd as administrator, go to `C:\Windows\System32\inetsrv`, and run `appcmd list wp`.

We can see the process ids and the corresponding site names.

Then set a breakpoint on the target function and refresh the page; it will break at the breakpoint.

2. ViewState basics

Before we try to exploit ViewState deserialization, we need to understand some related knowledge.

ASP.NET is a class library for developing Web applications, provided by Microsoft in the .NET Framework. It is packaged in the file `System.Web.dll`, exposes the `System.Web` namespace, and provides ASP.NET page processing, extensions, application and communication handling over the HTTP channel, and the infrastructure for Web Services.

In other words, ASP.NET is a Web library provided by the .NET Framework, and ViewState is a very distinctive feature provided by ASP.NET.

Why ViewState exists:

The HTTP model is stateless, which means that every time a client sends a request to the server to get a page, the server creates a new instance of the page class, and after one round trip that page instance is destroyed immediately. Suppose that while handling request `n+1` the server wants to use a value passed to it in request `n`; the page instance for request `n` was destroyed long ago, so where does it look for the value passed to the server last time? To meet this need, several state management techniques appeared, and ViewState is one of the state management techniques used by ASP.NET.

What does ViewState look like?

To understand ViewState, we must first know what a server control is.

In Microsoft's official naming, a ASP.NET page is called a Web Form. Besides distinguishing it from Windows Forms, the name also clearly describes its main purpose: "letting developers build Web pages the same way they build Windows Forms". So the functionality a ASP.NET Page has to provide must resemble a Windows Forms form: every Web Form must have a `< form runat="server" >< /form >` block, and all ASP.NET server controls

have to be placed inside that area, so that server controls such as ViewState can work smoothly.

Whether it is an HTML server control, a Web server control or a Validation server control, as long as it is an ASP.NET server control it must be placed inside the `< form runat="server" >< /form >` block, where the attribute `runat="server"` states that the form should be handled on the server side.

In its original state ViewState is a dictionary type. When responding with a page, ASP.NET serializes the state of all controls into a string, then inserts it into the page as the value of a hidden input and returns it to the client. When the client requests again, that hidden input passes the ViewState back to the server, the server deserializes the ViewState, obtains the properties, and assigns the corresponding values to the controls.

The security of ViewState:

Back in 2010, Microsoft published an article in *MSDN Magazine* discussing the security of ViewState and some defensive measures. The article held that ViewState mainly faces two threats: information disclosure and tampering.

Information disclosure threat:

The original ViewState only base64 encoded the serialized binary data, without using any kind of cryptographic algorithm to encrypt it, and it could easily be decoded and deserialized with `LosFormatter` (which has now been replaced by `ObjectStateFormatter`).

```
LosFormatter formatter = new LosFormatter();
object viewstateObj =
    formatter.Deserialize("/wEPDwULLTE2MTY2ODcyMjkPFgIeCHBhc3N3b3JkBQlzd29yZGZpc2hkZA==");
```

The result of deserialization is in fact a set of `System.Web.UI.Pair` objects.

To make sure ViewState does not leak information, ASP.NET 2.0 uses the `ViewStateEncryptionMode` property to enable encryption of ViewState. This property can be enabled through a page directive or in the application's `web.config` file.

```
<%@ Page ViewStateEncryptionMode="Always" %>
```

`ViewStateEncryptionMode` has three possible values: `Always`, `Never`, `Auto`

Tampering threat:

Encryption cannot prevent tampering: even with encrypted data, an attacker may still flip bits in the ciphertext. So data integrity techniques must be used to reduce the tampering threat, that is, a hash algorithm is used to create a message authentication code (MAC) for the message. Data validation can be enabled in `web.config` through `EnableViewStateMac`.

```
<%@ Page EnableViewStateMac="true" %>
```

Note: starting from .NET 4.5.2 the ViewStateMac feature is forcibly enabled, which means that even if you set EnableViewStateMac to false you cannot disable ViewState validation. Security bulletin [KB2905247](#) (sent to all Windows machines through the patch program on Patch Tuesday in September 2014) makes ASP.NET ignore the EnableViewStateMac setting. Rough steps after ViewStateMac is enabled:

- (1) The state of the page and of all participating controls is collected into a state graph object.
- (2) The state graph is serialized into binary format
 - a. The key value is appended to the serialized byte array.
 - b. A cryptographic hash is computed for the new serialized byte array.
 - c. The hash is appended to the end of the serialized byte array.
- (3) The serialized byte array is encoded as a base-64 string.
- (4) The base-64 string is written into the __VIEWSTATE form value in the page.

Using ViewState for deserialization exploitation

The real problem with ViewState is its potential deserialization vulnerability risk. ViewState uses ObjectStateFormatter to deserialize; although ViewState takes encryption and signing security measures, once web.config leaks and we obtain the key and algorithm used for its encryption and signing, we can apply the same encryption and signing to an ObjectStateFormatter deserialization payload and then send it to the server. That way, when ASP.NET deserializes it, it decrypts and validates normally, then hands the payload to ObjectStateFormatter to deserialize, triggering its deserialization vulnerability and achieving RCE.

3. The ViewState configuration in web.config

ASP.NET configures a website through web.config.

In web.config the following parameters can be used to turn some ViewState features on or off:

```
<pages enableViewState="false" enableViewStateMac="false"
viewStateEncryptionMode="Always" />
```

enableViewState: used to set whether viewState is enabled. But note that, as stated in **security bulletin KB2905247**, even if enableViewState is set to false in web.config, the ASP.NET server always parses ViewState passively. In other words, this option can affect the generation of ViewState, but not the passive parsing of ViewState. In fact **viewStateEncryptionMode** has a similar characteristic.

enableViewStateMac: used to set whether the ViewState Mac (validation) feature is enabled. Before **security bulletin KB2905247**, that is, before 4.5.2, setting this option to

false could disable the Mac validation feature. But after 4.5.2 ViewState Mac validation is forcibly enabled, because disabling this option brings serious security problems. Still, we can disable Mac validation by configuring the registry or by adding a dangerous setting in web.config; see the analysis later for details.

viewStateEncryptionMode: used to set whether the ViewState Encrypt (encryption) feature is enabled. This option has three possible values: Always, Auto, Never.

- Always means ViewState is always encrypted;
- Auto means that if a control requests encryption by calling the RegisterRequiresViewStateEncryption() method, the view state information will be encrypted; this is the default value;
- Never means the view state information is never encrypted, even if a control requested it.

In actual debugging I found that viewStateEncryptionMode affects the generation of ViewState, but when parsing a ViewState submitted by the client it is not this setting that decides whether to decrypt. See the analysis later for details.

In web.config the machineKey section further configures the validation and encryption features:

```
<machineKey validationKey="[String]" decryptionKey="[String]" validation="[SHA1 | MD5 | 3DES | AES | HMACSHA256 | HMACSHA384 | HMACSHA512 | alg:algorithm_name]" decryption="[Auto | DES | 3DES | AES | alg:algorithm_name]" />
```

Example:

```
<machineKey  
validationKey="BF579EF0E9F0C85277E75726BFC9D0260FADE8DE2864A583484AA132944F602D"  
decryptionKey="51FE611365277B07911521B7CAFE3766751D16C33D96242F0E63E93FB102BCE2"  
validation="HMACSHA256" />
```

Here **validationKey** and **decryptionKey** are the keys used for validation and encryption respectively, and **validation** and **decryption** are the algorithms used for validation and encryption (they can be omitted, in which case the default algorithm is used). The validation algorithms include SHA1, MD5, 3DES, AE, HMACSHA256, HMACSHA384, HMACSHA512. The encryption algorithms include DES, 3DES, AES. Because web.config is kept on the server, the security of ViewState is guaranteed as long as machineKey does not leak.

Now that we understand some of the ViewState configuration, let us look at how the .NET Framework actually handles the generation and parsing of ViewState.

4. The generation and parsing flow of ViewState

From some prior knowledge we know that ViewState uses the **Serialize** and **Deserialize** of ObjectStateFormatter to do the serialization and deserialization of ViewState. (LosFormatter is also used to serialize ViewState, but it has now been replaced by ObjectStateFormatter. LosFormatter's Serialize directly calls ObjectStateFormatter's Serialize.)

ObjectStateFormatter lives in the System.Web.UI namespace. Let us set a breakpoint on its Serialize function (there are several overloaded Serialize functions, be careful which one). Debug with dnspy, and after it breaks look at the stack backtrace:

From the backtrace we can clearly see that the Page class enters ObjectStateFormatter's Serialize function by calling SaveAllState.

4.1 The Serialize flow

Look at the code of the Serialize function (here I use the .Net 4.8 source, which has comments and is clearer):

```

private string Serialize(object stateGraph, Purpose purpose) {
    string result = null;

    MemoryStream ms = GetMemoryStream();
    try {
        Serialize(ms, stateGraph);
        ms.SetLength(ms.Position);

        byte[] buffer = ms.GetBuffer();
        int length = (int)ms.Length;

#if !FEATURE_PAL // FEATURE_PAL does not enable cryptography
        // We only support serialization of encrypted or encoded data through our
        internal Page constructors

        if (AspNetCryptoServiceProvider.Instance.IsDefaultProvider &&
            !_forceLegacyCryptography) {
            // If we're configured to use the new crypto providers, call into them if
            encryption or signing (or both) is requested.
            ...
        }
        else {
            // Otherwise go through legacy crypto mechanisms
#pragma warning disable 618 // calling obsolete methods
            if (_page != null && _page.RequiresViewStateEncryptionInternal) {
                buffer = MachineKeySection.EncryptOrDecryptData(true, buffer,
                    GetMacKeyModifier(), 0, length);
                length = buffer.Length;
            }
            // We need to encode if the page has EnableViewStateMac or we got passed in
            some mac key string
            else if ((_page != null && _page.EnableViewStateMac) || _macKeyBytes != null)
            {
                buffer = MachineKeySection.GetEncodedData(buffer, GetMacKeyModifier(), 0,
                    ref length);
            }
#pragma warning restore 618 // calling obsolete methods
        }

#endif // !FEATURE_PAL
        result = Convert.ToBase64String(buffer, 0, length);
    }
    finally {
        ReleaseMemoryStream(ms);
    }
}

```

```

    }
    return result;
}

```

At the start of the function it calls another overloaded `Serialize` function, whose job is to serialize the `stateGraph` into binary data:

```

MemoryStream ms = GetMemoryStream();
try {
    Serialize(ms, stateGraph);
    ms.SetLength(ms.Position);
    ...
}

```

After that it goes into the else branch:

```

if (_page != null && _page.RequiresViewStateEncryptionInternal) {
    buffer = MachineKeySection.EncryptOrDecryptData(true, buffer, GetMacKeyModifier(), 0,
length);
    length = buffer.Length;
}
// We need to encode if the page has EnableViewStateMac or we got passed in some mac key
string
else if ((_page != null && _page.EnableViewStateMac) || _macKeyBytes != null) {
    buffer = MachineKeySection.GetEncodedData(buffer, GetMacKeyModifier(), 0, ref
length);
}

```

There are two important flags here, `_page.RequiresViewStateEncryptionInternal` and `_page.EnableViewStateMac`. These two flags decide whether the serialized Binary data goes into the **`MachineKeySection.EncryptOrDecryptData()`** function or the **`MachineKeySection.GetEncodedData()`** function.

Of these, the `EncryptOrDecryptData()` function is used to encrypt and optionally to sign (validate), while `GetEncodedData()` is only used to sign (validate). We will analyse these two functions in detail shortly; first let us study these two flags.

These two flags decide what security measures the ViewState produced by the server takes. This matches the roles of `EnableViewStateMac` and `viewStateEncryptionMode` in `web.config` described earlier.

`_page.RequiresViewStateEncryptionInternal` comes from here:

```
internal bool RequiresViewStateEncryptionInternal {
    get {
        return ViewStateEncryptionMode == ViewStateEncryptionMode.Always ||
            _viewStateEncryptionRequested && ViewStateEncryptionMode ==
ViewStateEncryptionMode.Auto;
    }
}
```

The ViewStateEncryptionMode in it should come directly from web.config. So whether MachineKeySection.EncryptOrDecryptData is entered depends on the configuration in web.config. (Note that entering that function not only encrypts, it also signs.)

_page.EnableViewStateMac comes from here:

```
public bool EnableViewStateMac {
    get { return _enableViewStateMac; }
    set {
        // DevDiv #461378: EnableViewStateMac=false can lead to remote code execution, so
we
        // have an mechanism that forces this to keep its default value of 'true'. We
only
        // allow actually setting the value if this enforcement mechanism is inactive.
        if (!EnableViewStateMacRegistryHelper.EnforceViewStateMac) {
            _enableViewStateMac = value;
        }
    }
}
```

The corresponding field _enableViewStateMac is set to the default value true in the Page class's initialization function:

```
public Page() {
    _page = this; // Set the page to ourselves

    _enableViewStateMac = EnableViewStateMacDefault;
    ...
}
```

So whether _enableViewStateMac is modified depends on EnableViewStateMacRegistryHelper.EnforceViewStateMac.

Looking at the EnableViewStateMacRegistryHelper class, it has the following comment for EnforceViewStateMac:

```
// Returns 'true' if the EnableViewStateMac patch (DevDiv #461378) is enabled,  
// meaning that we always enforce EnableViewStateMac=true. Returns 'false' if  
// the patch hasn't been activated on this machine.  
public static readonly bool EnforceViewStateMac;
```

In other words: when the EnableViewStateMac patch is enabled, EnforceViewStateMac returns true, which means the EnableViewStateMac flag mentioned earlier always keeps its default value true.

The initialization function of the EnableViewStateMacRegistryHelper class shows further what EnforceViewStateMac is modified according to:

```
static EnableViewStateMacRegistryHelper() {  
    // If the reg key is applied, change the default values.  
    bool regKeyIsActive = IsMacEnforcementEnabledViaRegistry();  
    if (regKeyIsActive) {  
        EnforceViewStateMac = true;  
        SuppressMacValidationErrorsFromCrossPagePostbacks = true;  
    }  
  
    // Override the defaults with what the developer specified.  
    if (AppSettings.AllowInsecureDeserialization.HasValue) {  
        EnforceViewStateMac = !AppSettings.AllowInsecureDeserialization.Value;  
  
        // Exception: MAC errors from cross-page postbacks should be suppressed  
        // if either the <appSettings> switch is set or the reg key is set.  
        SuppressMacValidationErrorsFromCrossPagePostbacks |=  
        !AppSettings.AllowInsecureDeserialization.Value;  
    }  
    ...  
}
```

You can see that EnforceViewStateMac is modified in two cases:

- According to the IsMacEnforcementEnabledViaRegistry() function. That function takes a value from the registry; if that entry is 0, it means the EnableViewStateMac patch is disabled.

```
private static bool IsMacEnforcementEnabledViaRegistry() {
    try {
        string keyName = String.Format(CultureInfo.InvariantCulture,
@"HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\ .NETFramework\v{0}",
Environment.Version.ToString(3));
        int rawValue = (int)Registry.GetValue(keyName, "AspNetEnforceViewStateMac",
defaultValue: 0 /* disabled by default */);
        return (rawValue != 0);
    }
    catch {
        // If we cannot read the registry for any reason, fail safe and assume
        enforcement is enabled.
        return true;
    }
}
```

- According to `AppSettings.AllowInsecureDeserialization.HasValue`. That value should come from a dangerous setting in `web.config`:

```
<configuration>
...
  <appSettings>
    <add key="aspnet:AllowInsecureDeserialization" value="true" />
  </appSettings>
</configuration>
```

In summary, `ViewStateMac` is forcibly enabled by default. To turn that feature off, you must disable the `EnableViewStateMac` patch through the registry or through a dangerous setting in `web.config`.

4.2 The Deserialize flow

Look at the code of the `Deserialize` function:

```

private object Deserialize(string inputString, Purpose purpose) {
    if (String.IsNullOrEmpty(inputString)) {
        throw new ArgumentNullException("inputString");
    }

    byte[] inputBytes = Convert.FromBase64String(inputString);
    int length = inputBytes.Length;

    #if !FEATURE_PAL // FEATURE_PAL does not enable cryptography
        try {
            if (AspNetCryptoServiceProvider.Instance.IsDefaultProvider &&
                !_forceLegacyCryptography) {
                // If we're configured to use the new crypto providers, call into them if
                // encryption or signing (or both) is requested.
                ...
            }
            else {
                // Otherwise go through legacy crypto mechanisms
                #pragma warning disable 618 // calling obsolete methods
                if (_page != null && _page.ContainsEncryptedViewState) {
                    inputBytes = MachineKeySection.EncryptOrDecryptData(false, inputBytes,
                        GetMacKeyModifier(), 0, length);
                    length = inputBytes.Length;
                }
                // We need to decode if the page has EnableViewStateMac or we got passed in
                // some mac key string
                else if ((_page != null && _page.EnableViewStateMac) || _macKeyBytes != null)
                {
                    inputBytes = MachineKeySection.GetDecodedData(inputBytes,
                        GetMacKeyModifier(), 0, length, ref length);
                }
                #pragma warning restore 618 // calling obsolete methods
            }
        }
        catch {
            // MSRC 10405: Don't propagate inner exceptions, as they may contain sensitive
            // cryptographic information.
            PerfCounters.IncrementCounter(AppPerfCounter.VIEWSTATE_MAC_FAIL);
            ViewStateException.ThrowMacValidationError(null, inputString);
        }
    #endif // !FEATURE_PAL
    object result = null;
    MemoryStream objectStream = GetMemoryStream();
    try {

```

```

        objectStream.Write(inputBytes, 0, length);
        objectStream.Position = 0;
        result = Deserialize(objectStream);
    }
    finally {
        ReleaseMemoryStream(objectStream);
    }
    return result;
}

```

The important part is again the else branch inside it:

```

else {
    // Otherwise go through legacy crypto mechanisms
    if (_page != null && _page.ContainsEncryptedViewState) {
        inputBytes = MachineKeySection.EncryptOrDecryptData(false, inputBytes,
GetMacKeyModifier(), 0, length);
        length = inputBytes.Length;
    }
    // We need to decode if the page has EnableViewStateMac or we got passed in some mac
key string
    else if ((_page != null && _page.EnableViewStateMac) || _macKeyBytes != null) {
        inputBytes = MachineKeySection.GetDecodedData(inputBytes, GetMacKeyModifier(), 0,
length, ref length);
    }
}
}

```

Here a new flag `_page.ContainsEncryptedViewState` appears, used to decide whether to enter the `MachineKeySection.EncryptOrDecryptData()` function to decrypt. Look at where `ContainsEncryptedViewState` comes from:

```

if (_requestValueCollection != null) {

    // Determine if viewstate was encrypted.
    if (_requestValueCollection[ViewStateEncryptionID] != null) {
        ContainsEncryptedViewState = true;
    }
    ...
}

```

The comment shows that this flag is indeed used to decide whether the received viewstate is encrypted. Look at the result of reverse engineering with dnspy and it becomes even clearer:

That "__VIEWSTATEENCRYPTED" looks a lot like a field submitted in the request. Search for it, and indeed it is.

Looking at a request with encryption enabled, there really is such a field with no value:

So when ASP.NET parses ViewState, it does not decide whether the ViewState is encrypted from web.config, but from whether the request contains the __VIEWSTATEENCRYPTED field. In other words, even if we set Always in web.config, the server will still passively parse a ViewState that is only signed. (In the blog of the author of the ViewState plugin of the YsoSerial.NET tool I read that after .net 4.5 the encryption algorithm and key are needed. But I do not understand why, and in actual testing they do not seem to be needed either.)

5. The GetEncodedData signing function

The GetEncodedData() function is used to sign the serialized Binary data, for integrity validation. Look at its code (.NET 4.8):

```

// NOTE: When encoding the data, this method *may* return the same reference to the input
"buf" parameter
// with the hash appended in the end if there's enough space. The "length" parameter
would also be
// appropriately adjusted in those cases. This is an optimization to prevent unnecessary
copying of
// buffers.
[Obsolete(OBSOLETE_CRYPTO_API_MESSAGE)]
internal static byte[] GetEncodedData(byte[] buf, byte[] modifier, int start, ref int
length)
{
    EnsureConfig();

    byte[] bHash = HashData(buf, modifier, start, length);
    byte[] returnBuffer;

    if (buf.Length - start - length >= bHash.Length)
    {
        // Append hash to end of buffer if there's space
        Buffer.BlockCopy(bHash, 0, buf, start + length, bHash.Length);
        returnBuffer = buf;
    }
    else
    {
        returnBuffer = new byte[length + bHash.Length];
        Buffer.BlockCopy(buf, start, returnBuffer, 0, length);
        Buffer.BlockCopy(bHash, 0, returnBuffer, length, bHash.Length);
        start = 0;
    }
    length += bHash.Length;

    if (s_config.Validation == MachineKeyValidation.TripleDES || s_config.Validation ==
MachineKeyValidation.AES) {
        returnBuffer = EncryptOrDecryptData(true, returnBuffer, modifier, start, length,
true);
        length = returnBuffer.Length;
    }
    return returnBuffer;
}

```

Rough flow:

- The HashData() function computes the hash value.

- Check whether the original buffer is long enough; if it is, the hash value is appended right after data in the original buffer; otherwise a new buffer is allocated and data and the hash value are copied into it.
- Check whether the hash algorithm is 3DES or AES; if it is, call the EncryptOrDecryptData() function.

Let us first look at the HashData function:

```
internal static byte[] HashData(byte[] buf, byte[] modifier, int start, int length)
{
    EnsureConfig();

    if (s_config.Validation == MachineKeyValidation.MD5)
        return HashDataUsingNonKeyedAlgorithm(null, buf, modifier, start, length,
s_validationKey);
    if (_UseHMACSHA) {
        byte [] hash = GetHMACSHA1Hash(buf, modifier, start, length);
        if (hash != null)
            return hash;
    }
    if (_CustomValidationTypeIsKeyed) {
        return
HashDataUsingKeyedAlgorithm(KeyedHashAlgorithm.Create(_CustomValidationName),
                                buf, modifier, start, length,
s_validationKey);
    } else {
        return
HashDataUsingNonKeyedAlgorithm(HashAlgorithm.Create(_CustomValidationName),
                                buf, modifier, start, length,
s_validationKey);
    }
}
```

There are a few special flags here: `s_config.Validation`, `_UseHMACSHA`, `_CustomValidationTypeIsKeyed`, used to decide which function is entered to generate the hash.

`s_config.Validation` should be the signing algorithm set in `web.config`.

The other two flags come from the initialization settings made in the `InitValidationAndEncryptionSizes()` function according to the signing algorithm:

```

private void InitValidationAndEncryptionSizes()
{
    _CustomValidationName = ValidationAlgorithm;
    _CustomValidationTypeIsKeyed = true;
    switch(ValidationAlgorithm)
    {
        case "AES":
        case "3DES":
            _UseHMACSHA = true;
            _HashSize = SHA1_HASH_SIZE;
            _AutoGenValidationKeySize = SHA1_KEY_SIZE;
            break;
        case "SHA1":
            _UseHMACSHA = true;
            _HashSize = SHA1_HASH_SIZE;
            _AutoGenValidationKeySize = SHA1_KEY_SIZE;
            break;
        case "MD5":
            _CustomValidationTypeIsKeyed = false;
            _UseHMACSHA = false;
            _HashSize = MD5_HASH_SIZE;
            _AutoGenValidationKeySize = MD5_KEY_SIZE;
            break;
        case "HMACSHA256":
            _UseHMACSHA = true;
            _HashSize = HMACSHA256_HASH_SIZE;
            _AutoGenValidationKeySize = HMACSHA256_KEY_SIZE;
            break;
        case "HMACSHA384":
            _UseHMACSHA = true;
            _HashSize = HMACSHA384_HASH_SIZE;
            _AutoGenValidationKeySize = HMACSHA384_KEY_SIZE;
            break;
        case "HMACSHA512":
            _UseHMACSHA = true;
            _HashSize = HMACSHA512_HASH_SIZE;
            _AutoGenValidationKeySize = HMACSHA512_KEY_SIZE;
            break;
        default:
            ...
    }
}

```

You can see that only the MD5 signing algorithm sets `_UseHMACSHA` to false; every other algorithm sets it to true. Besides that, `_HashSize` is also set to the corresponding hash length according to the signing algorithm. So when computing an MD5 hash it enters the

HashDataUsingNonKeyedAlgorithm() function, and when computing the hash of other algorithms it enters the GetHMACSHA1Hash() function.

Let us first look at the HashDataUsingNonKeyedAlgorithm() function that is entered when the MD5 signing algorithm is used:

```
private static byte[] HashDataUsingNonKeyedAlgorithm(HashAlgorithm hashAlgo, byte[] buf,
byte[] modifier,
int start, int length, byte[]
validationKey)
{
    int totalLength = length + validationKey.Length + ((modifier != null) ?
modifier.Length : 0);
    byte [] bAll = new byte[totalLength];

    Buffer.BlockCopy(buf, start, bAll, 0, length);
    if (modifier != null) {
        Buffer.BlockCopy(modifier, 0, bAll, length, modifier.Length);
    }
    Buffer.BlockCopy(validationKey, 0, bAll, length, validationKey.Length);
    if (hashAlgo != null) {
        return hashAlgo.ComputeHash(bAll);
    } else {
        byte[] newHash = new byte[MD5_HASH_SIZE];
        int hr = UnsafeNativeMethods.GetSHA1Hash(bAll, bAll.Length, newHash,
newHash.Length);
        Marshal.ThrowExceptionForHR(hr);
        return newHash;
    }
}
```

Where the modifier here comes from we will discuss later; its length is generally 4 bytes. The flow of the HashDataUsingNonKeyedAlgorithm() function is as follows:

- Allocate a new block of memory whose length is data length + validationkey.length + modifier.length
- Copy data, modifier and validationkey into the newly allocated memory. What is special is that both modifier and validationkey are copied starting right next to data, which causes validationkey to overwrite modifier. So the real memory layout is: data + validationkey + '\x00'*modifier.length
- Set the hash length according to the MD5 algorithm, that is newHash. On this point, the code contains the hash length settings produced by the various algorithms:

```

private const int MD5_KEY_SIZE           = 64;
private const int MD5_HASH_SIZE         = 16;
private const int SHA1_KEY_SIZE         = 64;
private const int HMACSHA256_KEY_SIZE   = 64;
private const int HMACSHA384_KEY_SIZE   = 128;
private const int HMACSHA512_KEY_SIZE   = 128;
private const int SHA1_HASH_SIZE        = 20;
private const int HMACSHA256_HASH_SIZE  = 32;
private const int HMACSHA384_HASH_SIZE  = 48;
private const int HMACSHA512_HASH_SIZE  = 64;

```

The hash lengths corresponding to the various algorithms are MD5:16 SHA1:20 MACSHA256:32 HMACSHA384:48 HMACSHA512:64, all different.

- Call the UnsafeNativeMethods.GetSHA1Hash() function to compute the hash. That function is imported from webengine4.dll. Seeing this for the first time, I had some doubts: why does the MD5 algorithm call the GetSHA1Hash function? Let us keep that question for now. First let us see how the other algorithms generate their hashes.

When computing the hash of other algorithms, a hand written GetHMACSHA1Hash() function is called, implemented as follows:

```

private static byte[] GetHMACSHA1Hash(byte[] buf, byte[] modifier, int start, int length)
{
    if (start < 0 || start > buf.Length)
        throw new ArgumentException(SR.GetString(SR.InvalidArgumentValue, "start"));
    if (length < 0 || buf == null || (start + length) > buf.Length)
        throw new ArgumentException(SR.GetString(SR.InvalidArgumentValue, "length"));
    byte[] hash = new byte[_HashSize];
    int hr = UnsafeNativeMethods.GetHMACSHA1Hash(buf, start, length,
                                                    modifier, (modifier == null) ? 0 :
modifier.Length,
                                                    s_inner, s_inner.Length, s_outer,
s_outer.Length,
                                                    hash, hash.Length);

    if (hr == 0)
        return hash;
    _UseHMACSHA = false;
    return null;
}

```

You can see that internally it directly calls the UnsafeNativeMethods.GetHMACSHA1Hash() function, which is also a function imported from webengine4.dll. Just as when we looked at generating the MD5 hash value, the same question arises: why GetHMACSHA1Hash? Why do several algorithms all enter this one function? From the characteristics of their

arguments, and because we saw earlier that the hash lengths generated by the various algorithms differ, we can guess that perhaps the function internally chooses which algorithm to use based on the hash length.

Drag webengine4.dll into ida. Look at the GetSHA1Hash() function and the GetHMACSHA1Hash() function; their characteristics are as follows:

GetHMACSHA1Hash:

Both enter the GetAlgorithmBasedOnHashSize() function, so our guess was right: the algorithm really is chosen by the hash length.

6. The EncryptOrDecryptData encryption and decryption function

We saw earlier that both when encryption is enabled and when the AES\3DES signing algorithm is used, the MachineKeySection.EncryptOrDecryptData() function is entered. So what is the flow inside that function?

Let us first look at that function's declaration and comment:

```

internal static byte[] EncryptOrDecryptData(bool fEncrypt, byte[] buf, byte[] modifier,
int start, int length, bool useValidationSymAlgo, bool useLegacyMode, IVType ivType, bool
signData)

/* This algorithm is used to perform encryption or decryption of a buffer, along with
optional signing (for encryption)
* or signature verification (for decryption). Possible operation modes are:
*
* ENCRYPT + SIGN DATA (fEncrypt = true, signData = true)
* Input: buf represents plaintext to encrypt, modifier represents data to be appended to
buf (but isn't part of the plaintext itself)
* Output: E(iv + buf + modifier) + HMAC(E(iv + buf + modifier))
*
* ONLY ENCRYPT DATA (fEncrypt = true, signData = false)
* Input: buf represents plaintext to encrypt, modifier represents data to be appended to
buf (but isn't part of the plaintext itself)
* Output: E(iv + buf + modifier)
*
* VERIFY + DECRYPT DATA (fEncrypt = false, signData = true)
* Input: buf represents ciphertext to decrypt, modifier represents data to be removed
from the end of the plaintext (since it's not really plaintext data)
* Input (buf): E(iv + m + modifier) + HMAC(E(iv + m + modifier))
* Output: m
*
* ONLY DECRYPT DATA (fEncrypt = false, signData = false)
* Input: buf represents ciphertext to decrypt, modifier represents data to be removed
from the end of the plaintext (since it's not really plaintext data)
* Input (buf): E(iv + plaintext + modifier)
* Output: m
*
* The 'iv' in the above descriptions isn't an actual IV. Rather, if ivType =
IVType.Random, we'll prepend random bytes ('iv')
* to the plaintext before feeding it to the crypto algorithms. Introducing randomness
early in the algorithm prevents users
* from inspecting two ciphertexts to see if the plaintexts are related. If ivType =
IVType.None, then 'iv' is simply
* an empty string. If ivType = IVType.Hash, we use a non-keyed hash of the plaintext.
*
* The 'modifier' in the above descriptions is a piece of metadata that should be
encrypted along with the plaintext but
* which isn't actually part of the plaintext itself. It can be used for storing things
like the user name for whom this
* plaintext was generated, the page that generated the plaintext, etc. On decryption,
the modifier parameter is compared

```

```
* against the modifier stored in the crypto stream, and it is stripped from the message  
before the plaintext is returned.  
  
* In all cases, if something goes wrong (e.g. invalid padding, invalid signature,  
invalid modifier, etc.), a generic exception is thrown.  
  
*/
```

The comment says at the start that this function is used to encrypt/decrypt and optionally to sign/validate. There are four cases in total: encrypt+sign, encrypt only, decrypt+validate, decrypt only. The important ones are encrypt+sign and decrypt+validate.

- Encrypt+sign: fEncrypt = true, signData = true. Input: the original data to encrypt, modifier. Output: E(iv + buf + modifier) + HMAC(E(iv + buf + modifier)) (in the formula above E means encryption and HMAC means signing)
- Decrypt+validate: fEncrypt = false, signData = true. Input: the encrypted data to decrypt, modifier; buf is the E(iv + m + modifier) + HMAC(E(iv + m + modifier)) above. Output: m

Honestly, from the comment alone it seems we can already understand how this function encrypts and signs, and we could pick up python and learn to forge an encrypted viewstate (just kidding). Still, let us look at its code:

```
internal static byte[] EncryptOrDecryptData(bool fEncrypt, byte[] buf, byte[] modifier,  
int start, int length, bool useValidationSymAlgo, bool useLegacyMode, IVType ivType, bool  
signData)
```

The function has 9 parameters:

- The 1st parameter fEncrypt says whether this is encryption or decryption; true is encryption, false is decryption;
- The 2nd to 5th parameters buf, modifier, start, length relate to the original data;
- The 6th parameter useValidationSymAlgo says whether encryption uses the same algorithm as signing;
- The 7th parameter useLegacyMode relates to a custom algorithm and is generally false;
- The 8th parameter ivType relates to the initialization vector iv used in encryption; according to the comment the old IPType.Hash has been removed and IPType.Random is now used by default;
- The 9th parameter signData says whether to sign/validate.

There are some details worth mentioning about the 6th parameter useValidationSymAlgo:

We know that under the Serialize function there are two cases in which the EncryptOrDecryptData function is entered:

(1) Because encryption is enabled in the web.config configuration, the EncryptOrDecryptData() function is entered directly:

In this case EncryptOrDecryptData () has 5 parameters.

(2) After entering the GetEncodedData() function, because the AES/3DES signing algorithm is used, the EncryptOrDecryptData() function is entered again:

In this case EncryptOrDecryptData () has 6 parameters.

The two have different parameter counts, which shows that different overloads are entered.

Looking closely you find that when EncryptOrDecryptData () is entered because the AES/3DES signing algorithm was used, the 6th parameter useValidationSymAlgo is true. What is the point? Because entering GetEncodedData() first means the encryption feature is not enabled; at this point, because the AES/3DES signing algorithm is used, the EncryptOrDecryptData () function has to be entered again after signing. Entering EncryptOrDecryptData() requires deciding which encryption algorithm to use. So the 6th parameter is true, meaning encryption uses the same algorithm as signing. One more thing: in this case there are two signings, one inside GetEncodedData() and another after entering EncryptOrDecryptData() (as we will see later).

In the code below the operations related to decryption and validation are hidden; only the encryption and signing parts are shown.

```
// 541~543
System.IO.MemoryStream ms = new System.IO.MemoryStream();
ICryptoTransform cryptoTransform = GetCryptoTransform(fEncrypt, useValidationSymAlgo,
useLegacyMode);
CryptoStream cs = new CryptoStream(ms, cryptoTransform, CryptoStreamMode.Write);
```

This part first calls GetCryptoTransform to obtain the encryption tool, and then links the data stream to the encryption transform stream through CryptoStream. If you are not familiar with this process, see the [relevant Microsoft documentation](#).

The key point is how GetCryptoTransform() chooses the encryption tool. None of the function's 3 parameters seems to relate to an algorithm. Look at its code:

```
private static ICryptoTransform GetCryptoTransform(bool fEncrypt, bool
useValidationSymAlgo, bool legacyMode)
{
    SymmetricAlgorithm algo = (legacyMode ? s_oSymAlgoLegacy : (useValidationSymAlgo ?
s_oSymAlgoValidation : s_oSymAlgoDecryption));
    lock(algo)
        return (fEncrypt ? algo.CreateEncryptor() : algo.CreateDecryptor());
}
```

algo stands for the corresponding algorithm class, so the key parts are s_oSymAlgoValidation and s_oSymAlgoDecryption. Let us see where they come from:

The ConfigureEncryptionObject() function:

```
switch (Decryption)
{
    case "3DES":
        s_oSymAlgoDecryption = CryptoAlgorithms.CreateTripleDES();
        break;
    case "DES":
        s_oSymAlgoDecryption = CryptoAlgorithms.CreateDES();
        break;
    case "AES":
        s_oSymAlgoDecryption = CryptoAlgorithms.CreateAes();
        break;
    case "Auto":
        if (dKey.Length == 8) {
            s_oSymAlgoDecryption = CryptoAlgorithms.CreateDES();
        } else {
            s_oSymAlgoDecryption = CryptoAlgorithms.CreateAes();
        }
        break;
}

if (s_oSymAlgoDecryption == null) // Shouldn't happen!
    InitValidationAndEncryptionSizes();

switch(Validation)
{
    case MachineKeyValidation.TripleDES:
        if (dKey.Length == 8) {
            s_oSymAlgoValidation = CryptoAlgorithms.CreateDES();
        } else {
            s_oSymAlgoValidation = CryptoAlgorithms.CreateTripleDES();
        }
        break;
    case MachineKeyValidation.AES:
        s_oSymAlgoValidation = CryptoAlgorithms.CreateAes();
        break;
}
```

So it seems the corresponding encryption classes are already assigned when the website is initialized.

Continue looking at the code of EncryptOrDecryptData():

```

// Lines 545–579
// DevDiv Bugs 137864: Add IV to beginning of data to be encrypted.
// IVType.None is used by MembershipProvider which requires compatibility even in SP2
mode (and will set signData = false).
// MSRC 10405: If signData is set to true, we must generate an IV.
bool createIV = signData || ((ivType != IVType.None) && (CompatMode >
MachineKeyCompatibilityMode.Framework20SP1));

if (fEncrypt && createIV)
{
    int ivLength = (useValidationSymAlgo ? _IVLengthValidation : _IVLengthDecryption);
    byte[] iv = null;

    switch (ivType) {
        case IVType.Hash:
            // iv := H(buf)
            iv = GetIVHash(buf, ivLength);
            break;

        case IVType.Random:
            // iv := [random]
            iv = new byte[ivLength];
            RandomNumberGenerator.GetBytes(iv);
            break;
    }

    Debug.Assert(iv != null, "Invalid value for IVType: " + ivType.ToString("G"));
    cs.Write(iv, 0, iv.Length);
}

cs.Write(buf, start, length);
if (fEncrypt && modifier != null)
{
    cs.Write(modifier, 0, modifier.Length);
}

cs.FlushFinalBlock();
byte[] paddedData = ms.ToArray();

```

The start of this part generates the IV. The IV is the initialization vector used in encryption; it must be random, to prevent a repeated IV from letting the ciphertext be broken.

- ivLength is 64. Here 64 random bytes are generated as the iv.

- `cs.Write()` is called three times, writing `iv`, `buf` and `modifier` respectively. `cs` is the `CryptoStream` instance created earlier, used to route the data stream into the encryption stream. This matches the formula $E(iv + buf + modifier)$ we mentioned earlier.
- `ms.ToArray()` is called, which returns the byte sequence produced after encryption is complete.

Continue looking at the code of `EncryptOrDecryptData()`:

```
// Lines 550–644
// DevDiv Bugs 137864: Strip IV from beginning of unencrypted data
if (!fEncrypt && createIV)
{
    // strip off the first bytes that were random bits
    ...
}
else
{
    bData = paddedData;
}
...
// At this point:
// If fEncrypt = true (encrypting), bData := Enc(iv + buf + modifier)
// If fEncrypt = false (decrypting), bData := plaintext

if (fEncrypt && signData) {
    byte[] hmac = HashData(bData, null, 0, bData.Length);
    byte[] bData2 = new byte[bData.Length + hmac.Length];

    Buffer.BlockCopy(bData, 0, bData2, 0, bData.Length);
    Buffer.BlockCopy(hmac, 0, bData2, bData.Length, hmac.Length);
    bData = bData2;
}

// At this point:
// If fEncrypt = true (encrypting), bData := Enc(iv + buf + modifier) + HMAC(Enc(iv + buf
// + modifier))
// If fEncrypt = false (decrypting), bData := plaintext

// And we're done
return bData;
```

This is the last step: the byte sequence produced by the encryption is passed to `HashData`, which generates the hash value and appends it after the byte sequence.

This matches the earlier formula $E(iv + buf + modifier) + HMAC(E(iv + buf + modifier))$.

Having read the code of the `EncryptOrDecryptData()` function, we now understand its flow. Summed up it is really just one formula, yes: $E(iv + buf + modifier) + HMAC(E(iv + buf + modifier))$.

7. Where modifier comes from

In the signing and encryption processes above, a key variable called `modifier` was used; it is used together with the key for signing and encryption. This variable comes from the `GetMacKeyModifier()` function:

```

// This will return the MacKeyModifier provided in the LOSFormatter constructor or
// generate one from Page if EnableViewStateMac is true.
private byte[] GetMacKeyModifier() {
    if (_macKeyBytes == null) {
        // Only generate a MacKeyModifier if we have a page
        if (_page == null) {
            return null;
        }

        // Note: duplicated (somewhat) in GetSpecificPurposes, keep in sync

        // Use the page's directory and class name as part of the key (ASURT 64044)
        uint pageHashCode = _page.GetClientStateIdentifier();

        string viewStateUserKey = _page.ViewStateUserKey;
        if (viewStateUserKey != null) {
            // Modify the key with the ViewStateUserKey, if any (ASURT 126375)
            int count = Encoding.Unicode.GetByteCount(viewStateUserKey);
            _macKeyBytes = new byte[count + 4];
            Encoding.Unicode.GetBytes(viewStateUserKey, 0, viewStateUserKey.Length,
            _macKeyBytes, 4);

        }
        else {
            _macKeyBytes = new byte[4];
        }

        _macKeyBytes[0] = (byte)pageHashCode;
        _macKeyBytes[1] = (byte)(pageHashCode >> 8);
        _macKeyBytes[2] = (byte)(pageHashCode >> 16);
        _macKeyBytes[3] = (byte)(pageHashCode >> 24);
    }
    return _macKeyBytes;
}

```

The function's flow:

- At the start the function computes a pageHashCode through `_page.GetClientStateIdentifier`;
- If there is a viewStateUserKey, then modifier = pageHashCode + ViewStateUserKey;
- If there is no viewStateUserKey, then modifier = pageHashCode

First look at where pageHashCode comes from:

```
// This is a non-cryptographic hash code that can be used to identify which Page
generated
// a __VIEWSTATE field. It shouldn't be considered sensitive information since its inputs
// are assumed to be known by all parties.
internal uint GetClientStateIdentifier() {
    // Use non-randomized hash code algorithms instead of String.GetHashCode.

    // Use the page's directory and class name as part of the key (ASURT 64044)
    // We need to make sure that the hash is case insensitive, since the file system
    // is, and strange view state errors could otherwise happen (ASURT 128657)
    int pageHashCode = StringUtil.GetNonRandomizedHashCode(TemplateSourceDirectory,
ignoreCase:true);
    pageHashCode += StringUtil.GetNonRandomizedHashCode(GetType().Name, ignoreCase:true);

    return (uint)pageHashCode;
}
```

As the comment also shows, it computes the hash values of the directory and the class name, adds them and returns the result. So pageHashCode is 4 bytes. This means we can compute a page's pageHashCode by hand; directory and class name should be the site path and the site name respectively. Besides that, it can also be extracted from the hidden field "__VIEWSTATEGENERATOR" in the page, as shown below:

The relationship between "__VIEWSTATEGENERATOR" and pageHashCode is here:

Now look at where ViewStateUserKey comes from:

According to the official description, ViewStateUserKey is: assigning an identifier for an individual user in the ViewState variable associated with the current page.

So ViewStateUserKey is a random string value that must be tied to the user. If a site uses ViewStateUserKey, we should guess it from the SessionID or a cookie. In CVE-20202-0688, the SessionID is taken as the ViewStateUserKey.

8. Forging ViewState

After all the code pasting and analysis above, we now roughly understand how ASP.NET generates and parses ViewState. This helps us understand how to forge a ViewState. Of course, forging a ViewState still requires web.config to leak, so that its keys and algorithms are known.

- If the signing algorithm is not AES/3DES, then no matter whether encryption is enabled, we only need to generate a signed ViewState using its signing algorithm and key. Because the "__VIEWSTATEENCRYPTED" field is not used when that ViewState is sent, ASP.NET goes straight into GetDecodedData() for signature validation when parsing it, and no longer performs the decryption step.

- If the signing algorithm is AES/3DES, then no matter whether encryption is enabled, we only need to do as described earlier: sign the data once, encrypt it once, then sign it once more. Then send it to the server; ASP.NET enters GetDecodedData(), then first enters EncryptOrDecryptData() for one validation and decryption, and after coming out validates once more.

Put another way: whatever signing algorithm is used and whether or not encryption is enabled, when we forge a ViewState we forge it following the normal steps for the case where encryption is not enabled.

9. Appendix:

[1] ysoserial.net

<https://github.com/pwntester/ysoserial.net>

[2] viwgen (a viewstate generation tool written in python; it does not depend on .NET, which makes it convenient for automation scripts)

<https://github.com/0xacb/viewgen>

[3] What View State is and how it works in ASP.NET

<https://www.c-sharpcorner.com/UploadFile/225740/what-is-view-state-and-how-it-works-in-Asp-Net53/>

[4] Official Microsoft documentation: ASP.NET server controls overview

<https://docs.microsoft.com/zh-cn/troubleshoot/aspnet/server-controls>

[5] MSDN Magazine article: ViewState security

<https://docs.microsoft.com/en-us/archive/msdn-magazine/2010/july/security-briefs-view-state-security>

[6] Security bulletin KB2905247

<https://docs.microsoft.com/en-us/security-updates/SecurityAdvisories/2013/2905247?redirectedfrom=MSDN>

[7] Using ViewState

<http://appetere.com/post/working-with-viewstate>

[8] Exchange CVE-2020-0688

<https://www.thezdi.com/blog/2020/2/24/cve-2020-0688-remote-code-execution-on-microsoft-exchange-server-through-fixed-cryptographic-keys>

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